

Introduction to Machine Learning

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed.

Types of Machine Learning

1. Supervised Learning

Supervised learning uses labeled data to train a model.

Examples:

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest

Applications:

- House Price Prediction
- Spam Detection
- Credit Risk Assessment

2. Unsupervised Learning

Unsupervised learning works with unlabeled data.

Examples:

- K-Means Clustering
- Hierarchical Clustering
- Principal Component Analysis (PCA)

Applications:

- Customer Segmentation
- Market Basket Analysis

3. Reinforcement Learning

Reinforcement learning trains an agent through rewards and penalties.

Applications:

- Robotics

- Self-driving Cars
 - Game Playing
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Linear Regression

Linear Regression is used to predict continuous values.

Equation:

$$y = mx + b$$

Where:

- y = dependent variable
- x = independent variable
- m = slope
- b = intercept

Advantages:

- Easy to understand
- Fast to train

Disadvantages:

- Assumes linear relationship
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Decision Trees

Decision Trees split data into branches based on conditions.

Advantages:

- Easy to interpret
- Handles non-linear relationships

Disadvantages:

- Can overfit training data
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Model Evaluation Metrics

Regression Metrics

- MAE (Mean Absolute Error)
- MSE (Mean Squared Error)
- RMSE (Root Mean Squared Error)
- R^2 Score

Classification Metrics

- Accuracy
 - Precision
 - Recall
 - F1 Score
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Python Libraries for Machine Learning

NumPy

Used for numerical computations.

Pandas

Used for data manipulation and analysis.

Matplotlib

Used for data visualization.

Scikit-learn

Provides machine learning algorithms and utilities.

TensorFlow

Deep learning framework developed by Google.

PyTorch

Deep learning framework widely used in research and industry.

Interview Questions

1. What is Machine Learning?
2. Explain the difference between supervised and unsupervised learning.
3. What is overfitting?
4. What is underfitting?
5. Explain bias and variance.
6. What is cross-validation?
7. What is PCA?
8. What is feature engineering?
9. What is the difference between Random Forest and Decision Tree?
10. What evaluation metrics are used for classification problem